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Towards a Sustainable City Policy for Managing Shared Dockless E-Scooters

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Abstract

Based on the anonymized dataset of one million shared scooter rides collected by the Tel Aviv municipality, we investigate the behavior of the shared e-scooter users in Tel Aviv and the resulting dynamics of the scooters' spatio-temporal patterns. The users' choice of the shared e-scooters as a transportation mode follows fast and frugal heuristics: the user who activated the shared scooter's application continues with the ride when the closest available scooter is close enough, at up to 50-100 meters from her. Riders strongly prefer bike paths for riding and essentially deviate from the shortest path between the origin and destination and choose the routes with the higher percentage of bike paths. The longer the route, the higher the fraction of the bike paths in the rider's route and the difference between the shortest and chosen paths. We also demonstrate that the operators fail to match the supply to the demand and the scooters are oversupplied in the central part of the city, where the demand is high, and undersupplied in the rest of the city, where the demand is essentially lower. Based on the analysis we propose a new policy of spatial adjustment between the demand and supply.

Keywords: Micromobility; dockless e-scooters; user behavior; urban transport policy

1. Introduction

The history of the Shared Dockless E-Scooters (SDES) started in Singapore in 2016, just 8 years ago [1]. Now, at the start of 2024, almost a thousand cities in the world have introduced this novel shared transport mode and only a

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few have banned it after the introduction [2, 3]. Shared e-scooter trips mostly substitute public transport trips or walking and all over the world the average ride time is about 15-20 minutes, and the average trip length varies from 2 to 3 km. Worldwide, pedestrians and drivers are concerned about scooters riding on the sidewalk or against the traffic [4]. Cities want to make the SDES easily available to potential users while preserving safety and non-conflict interaction between the riders and non-riders. They thus seek a sustainable SDES policy that accounts for the interests of the SDES users and non-users on the one hand, while preserving operators' economic incentives for providing the service. Tel Aviv SDES policy is very simple: (a) The municipality has established 779 parking cells in the central and southern parts of the city (Figure 1a) and does not allow parking beyond the cells in these areas; (b) riding on the sidewalk is prohibited; (c) an operator can activate a fleet of 1800 vehicles, and must allocate at least 6% of its supply in the southern quarters 7 and 8 and at least 3% in the eastern quarter 9 (Figure 1a). Based on the Tel-Aviv SDES ridership we propose a policy update that will adjust the demand and supply in the urban space.

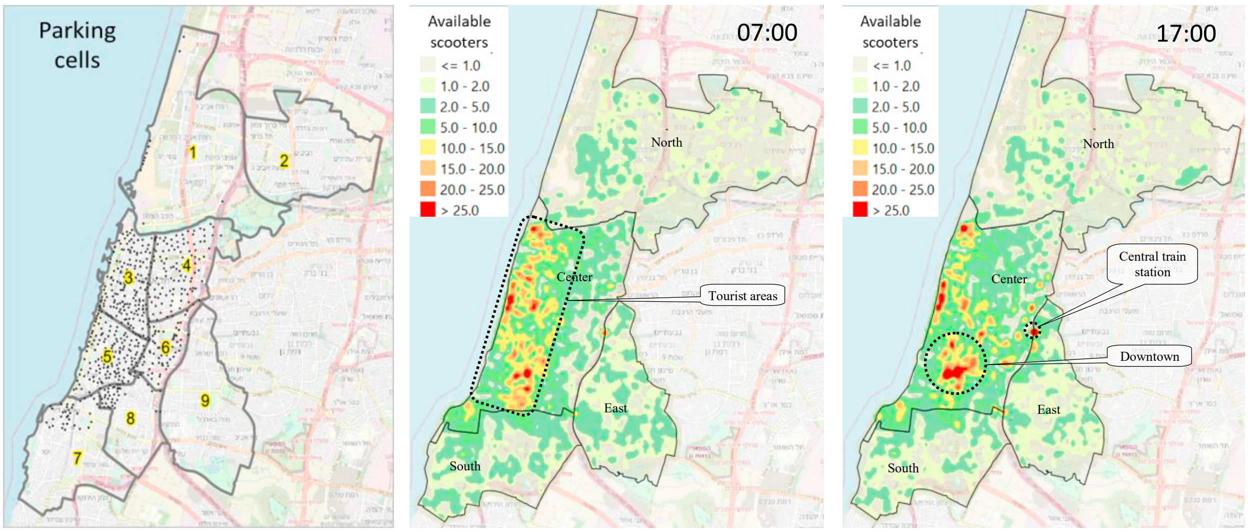
2. The Data

Our study is based on two large datasets. The first is the anonymized trip data on 1M rides collected by the Tel Aviv municipality using the POPULUS software [5, 6] during 22 working days of March 2023. The SDES potential users were 480K Tel Aviv residents and the same number of visitors. At that time, about 7500 e-scooters were operated by 5 companies in Tel Aviv (52 km^2 area). The data on scooters supplied by five city operators are combined. The routes of 10% of the trips have a continuous GPS presentation at a 10-second frequency. The second dataset is established by one of the operators and contains the time and location of 730K activations of the company's application, and, if the ride is chosen, the time, location, and plate number of the chosen scooter.

3. E-Scooter Users' Behavior and Operators' Actions

3.1. Spatial patterns of supply and demand

The users ride and leave scooters near the destination of their trips or at the closest parking cells, while operators charge and relocate scooters, mostly at night or during the day in response to real-time information on the high unsatisfied demand. The resulting distribution of scooters is very non-uniform - the supply is very high in several locations in the center of the city, and much lower at the periphery. Figures 1b and c present high supply in the tourist areas of Tel Aviv in the morning (7:00) and near the central train station at the end of the workday (17:00).



a b c

Figure 1. (a) Tel Aviv partition into Quarters and scooter parking cells in Tel Aviv on the base of city partition into Center, South, North, and East areas; (b) spatial pattern of supply at 07:00 and (c) at 17:00.

The spatial pattern of demand can be characterized by the average parking time of scooters. Figure 2a presents this pattern at a resolution of the city's Traffic Analysis Zones (TAZ). As can be seen, the demand is essentially heterogeneous and, basically, essentially higher in the center of the city than over three peripheral areas. Many scooters are used just after they are parked, while many scooters remain unused for half a day or even longer (Figure 2b). Parking near the bike paths decreases average parking time by almost an hour, especially in the central area, where the density of the cycling paths is the highest (Figure 2c).

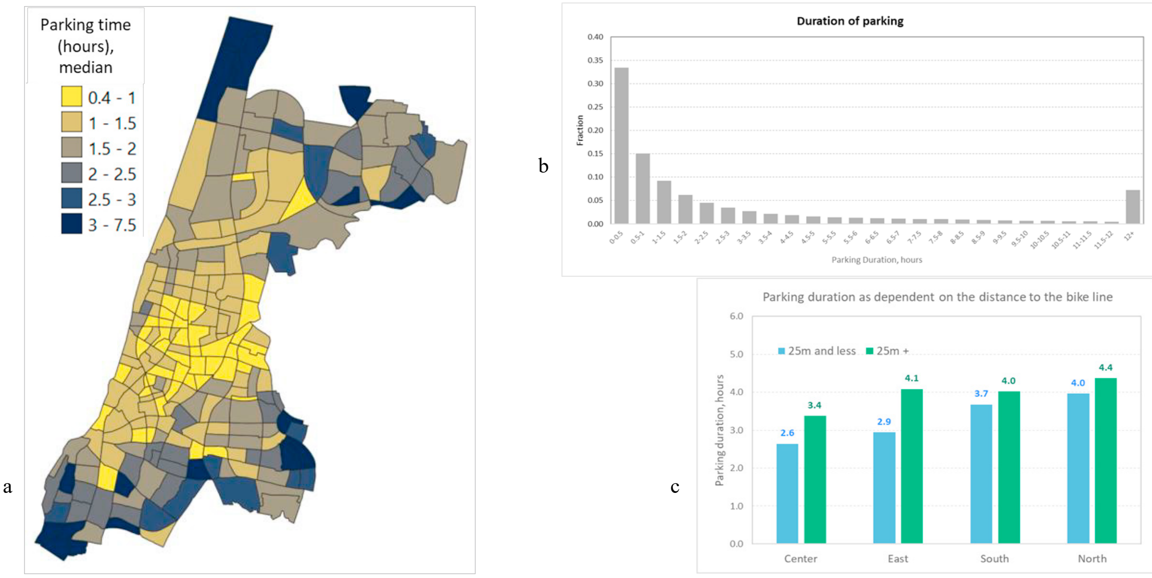


Figure 2. The average duration of parking during the active part of a day, 06:00-24:00, by Tel Aviv TAZ areas (a); The distribution of scooters' parking duration (b) and the average parking duration of the scooters that are located closer than 5 m to a bike path and further away.

The high heterogeneity of the demand results in the demand-supply match at some locations and mismatch at another (Figure 3). We hypothesize the Mohring-like feedback [7] as a reason for this phenomenon – the operators over-supply vehicles to the high-demand areas to guarantee a profit, and, given fleet limitation, lack scooters to allocate elsewhere, with the consequent under-supply and a subsequent decline in demand in the rest of the city. As mentioned, the city's SDES policy establishes quotes on scooters' allocation in peripheral areas only.

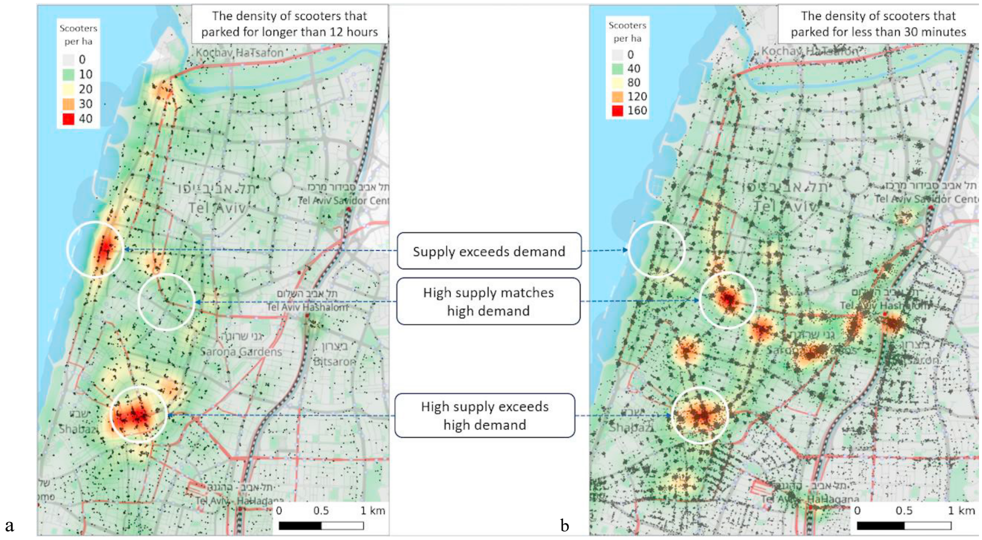


Figure 3. Averaged, over days, density of scooters that (a) parked for longer than 12 hours; and (b) parked for less than 30 minutes.

3.2. Riders' Route Choice

Riders' routes are, on average, 2.3 km, 20% longer than the average shortest path between the origin and the destination. About 90% of routes deviate from the shortest paths (Figure 4a). The view of individual trajectories (Figure 4b) suggests that riders prefer routes with a higher share of the bike paths - the share of segments with the bike path in the chosen route is indeed essentially higher than in the shortest path and grows with the increase in the route length (Figure 5a). The longer trip is balanced by the higher speed on the cycling path segments that, on average, is 20% higher than on those without (Figure 5b). The map of the scooterists' links usage (Figure 5c) provides a basis for the necessary future development of the cycling network planned before the era of the massive e-scooter usage.

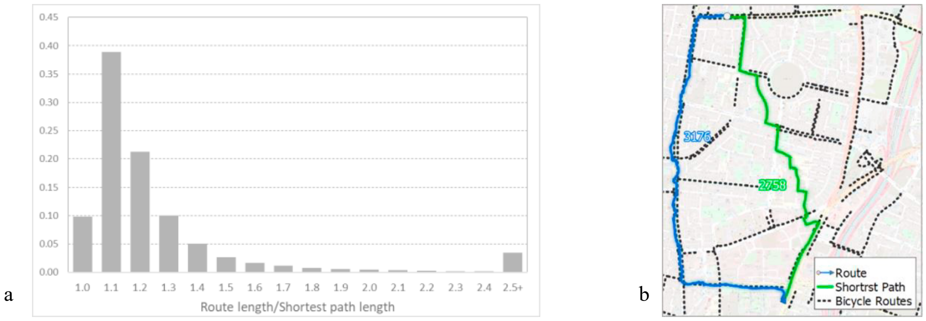


Figure 4. (a) The ratio of the actual route length to the shortest path between the rider's origin and destination; (b) Example of the chosen route and the shortest path between the rider's origin and destination.

Figure 5. (a) The share of the bike paths in the actual and shortest paths; (b) The riders' speeds on the bike - and non-bike-path-parts of the route: (c) Monthly utilization of street segments by riders in the Tel Aviv center

3.3. To Ride or Not to Ride?

The ride/no-ride decision after activating the operator's mobile application follows a "fast and frugal" [8] heuristic. If the decision is positive (roughly, 2/3 of cases, according to the available operator's data), then 95% of these users activate a scooter that is located closer than 110 m (Figure 6a). In 75% of cases, the activated scooter is the closest one (Figure 6b), and the longer the scooter is parked, the lower the probability of choosing it (Figure 6c). Compared to the activations that did not continue with a ride, the distance to the closest scooter in the vicinity of user's location is a major factor of the ride/no-ride decision (Figure 6d): In more than 30% of cases, the users who activated the application and found no scooters at a distance less than 100 m did not continue with a ride, and the probability that the no-ride user had a scooter at a distance less than 40 m is twice lower than for the ride-user, 40% vs. 80%. Spatially, activations almost always continue with a ride in the Tel Aviv center, while on the periphery the percentage of no-ride decisions reaches 15-25% (Figure 6e).

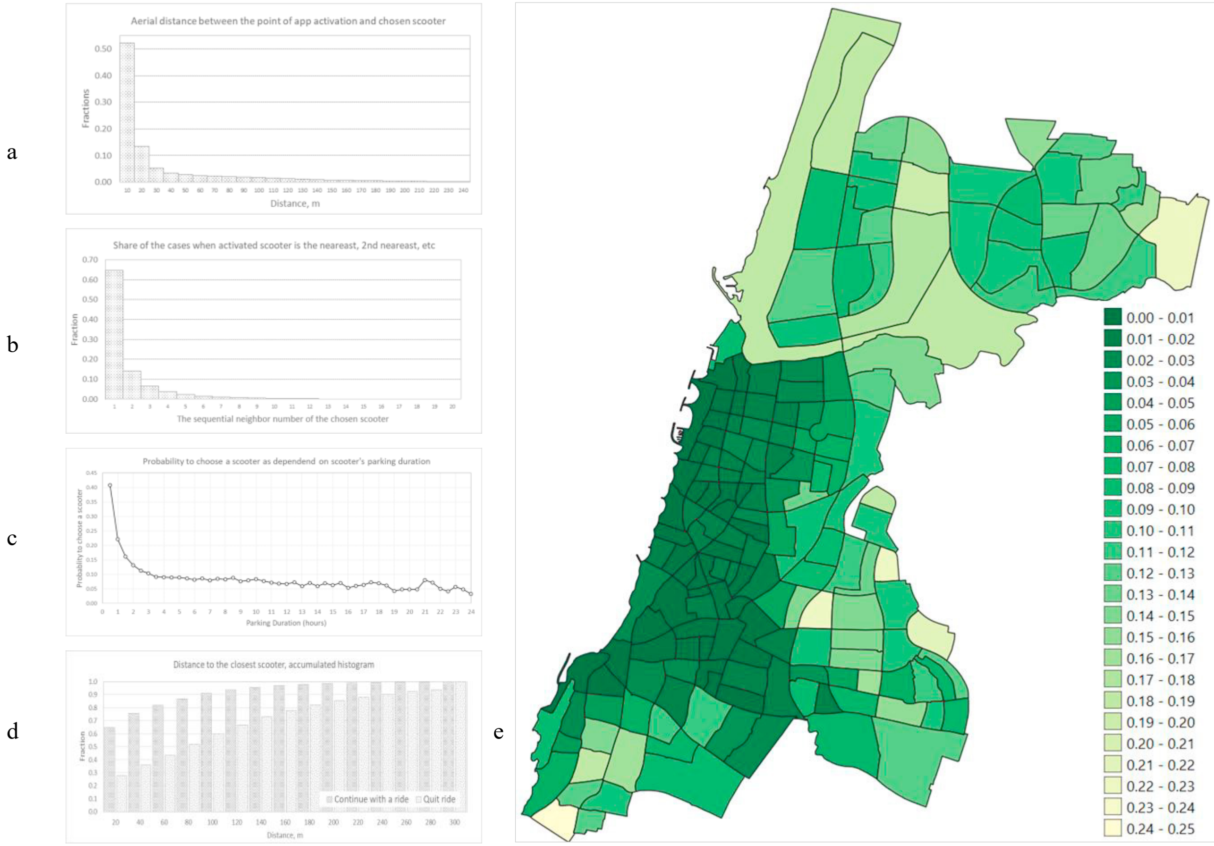


Figure 6. (a) The aerial distance between the point of application activation and location of the chosen scooter; (b) The share of cases where the chosen scooter is the nearest, second nearest, etc.; (c) Fraction of scooters chosen for ride among all scooters of the same parking duration with the radius of 150m aerial distance from the point application activation; (d) Distance to the closest scooter from the point of application activation for ride- and no-ride activations; (e) The share of no-ride application activations, at resolution of TAZ.

3.4. Policy Proposal

The current Tel Aviv SDES policy limits the operator's active fleet to 1800 vehicles and requests allocating 6% of the fleet in the South and 3% in the East of the city. We consider these limitations as a reason for oversupply in the center of the city and undersupply in the rest of the area and suggest an SDES-supervising policy that would enforce the match between the demand and supply all over the city and be a win-win for the operators and the city. The pre-condition for the new policy is establishing scooter parking cells all over the populated, or more general, attractive for riders, areas of the city and letting users leave a scooter at the parking cell only. The operators would thus know how many scooters are available everywhere in the city, while users could find a scooter easily. Based on the above analysis, parking cells must be established at a distance that is not higher than 200 m between each other.

Operators would now know parking cells where the supply essentially exceeds the demand and where the demand is not satisfied. Scooters from the oversupplied locations should be instantaneously relocated to the undersupplied ones, and, according to the above analysis, the scooters that should be chosen for this relocation are the ones with the longest parking time. The threshold parking time that makes a scooter candidate for relocation is a policy parameter. As the available data suggests, 8-12 hours of idle time looks like a reasonable option. Whether relocation should be performed instantaneously during operating hours or after that, depends on the traffic conditions and may be adjusted locally. These details should be established based on the deeper analysis that accounts for the origin-destination flows, which we did not consider in this short paper, and on the data on operators' economic and operational abilities. The natural choice of *where* long-not-used scooters should be relocated is the second policy parameter. It can be e.g., the closest undersupplied cell. The number of scooters available for relocation may be too high or, vice versa, insufficient to satisfy the existing demand. In this case, the operators' quota should be decreased or increased.

4. Conclusion

High-resolution analysis of the Tel Aviv shared e-scooter data exposes the basic features of the shared e-scooter users' behavior regarding ride/not-ride decisions after activating the operator's application and the choice of the route. The resulting dynamics of this inherently complex system are characterized by the essential mismatch between the demand and supply. Based on the high-resolution analysis of the scooterists' behavior and scooters' supply by the operators, we propose an SDES-supervising policy in the city that aims at establishing proper demand-supply balance and is a win-win for the citizens and visitors on the one hand and operators on the other. The policy's applicability and economic effectiveness still demand assessment within the broader framework that includes all three major system actors – the public, municipality, and operators.

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